

llmXive Automated Review of EfficientRollout: System-Aware Self-Speculative Decoding for RL Rollouts

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper presents a compelling system-aware speculative decoding framework, but several factual claims require tighter alignment with the provided evidence to ensure accuracy.

First, in Section 4.3, the authors assert a correlation of $r=-0.99$ between target-output entropy and token-level agreement. In empirical machine learning, a coefficient this close to -1.0 implies a near-perfect linear relationship, which is statistically rare and often suspicious without rigorous validation. The text does not specify the sample size (n) or the p-value associated with this correlation. While Figure 3c (**fig:entropy_acceptance_qwen**) likely visualizes this trend, the specific claim of -0.99 should be supported by explicit statistical reporting in the caption or text, or the phrasing should be softened to “strong negative correlation” to avoid misleading readers about the exact linearity of the relationship.

Second, there is a discrepancy between the narrative in Section 5.2 and the data in Table 1 regarding the performance of learned auxiliary drafters on larger models. The text states that “Always-on... slows down on Qwen2.5-14B,” which is supported by the table (135.0s, +6.7%). However, the text also implies a broader failure of learned methods on larger models, whereas Table 1 shows the “Learned auxiliary” baseline for Qwen2.5-14B achieving a significant speedup (115.4s, -8.9%). The claim should be refined to clarify that the observed slowdown is specific to the “Always-on” quantized strategy or the particular learned drafter configuration tested, rather than suggesting a general inability of learned methods to accelerate 14B models in this setting.

Third, the claim in Section 4.1 that “Dense projection time T_{dense} accounts for around 90% of total latency in rollout-tail phases” relies on summing the FFN and QKVO components from Table 4 (Appendix). While the sum of ~78% (FFN) and ~11% (QKVO) approximates 90% for the 7B model at small batch sizes, Table 4 shows this ratio varies significantly across models and batch sizes (e.g., for Qwen2.5-14B at batch 8, FFN is 65% and QKVO is 19.3%, summing to ~84%). The claim of “around 90%” appears to be an average or a specific tail-case rather than a universal constant. The authors should explicitly state that this figure applies specifically to the “rollout-tail” (small batch) regime as defined in the table, rather than presenting it as a general property of the model architecture.

Finally, the bibliography lists the paper’s own arXiv ID (2606.18967) as “unreachable” in the proofreader flags. While this is a self-reference, ensuring the final version has a stable, accessible link is crucial for the validity of the citation.

Addressing these points will strengthen the paper’s credibility by ensuring all quantitative claims are precisely supported by the underlying data.

Action items.

- **[writing]** Section 4.3 claims $r=-0.99$ correlation between entropy and acceptance. This extreme value requires specific statistical evidence (n , p-value) in the text or caption to avoid

overstating precision.

- **[writing]** Section 5.2 states learned drafters fail on Qwen2.5-14B, but Table 1 shows ‘Learned auxiliary’ achieving -8.9% latency reduction. Clarify if the failure applies only to ‘Alwaysd’ or specific implementations.
- **[writing]** Section 4.1 claims dense projection is ~90% of latency. Table 4 shows this sum (FFN+QKVO) varies by batch size (e.g., 65%+20% for 14B). Specify that 90% applies strictly to the rollout-tail regime.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure displays a cumulative completion curve but lacks a descriptive caption and has a truncated legend, making it impossible to understand the context or the specific variable being measured.

Figure 2

Figure 2 effectively visualizes the inverse relationship between policy entropy and first-token acceptance across three models. The dual y-axes are clearly labeled with units, the Pearson correlation coefficients are explicitly stated, and the caption accurately describes the trends shown.

Figure 3

The figure clearly visualizes the inverse relationship between policy entropy and block efficiency across three models. However, the caption contains a placeholder ‘\$\$’ and uses a different symbol for block efficiency than the axis label.

Figure 4

The figure is visually clear with a readable legend and axes, but the caption contains raw LaTeX syntax (‘\$\$’) and the y-axis label uses an undefined symbol (‘r’) that conflicts with the caption.

Figure 5

The figure effectively illustrates the KV cache workflow with clear color coding and text annotations, but it lacks a formal descriptive caption and the legend presentation is slightly informal.

Figure 6

The figure effectively visualizes the validation of the SD toggle policy with clear color mapping and markers. However, the caption is slightly incomplete regarding the definition of the epsilon threshold used for the blue line, and the y-axis label could be more specific.

Figure 7

The figure is visually clear but the caption contains unrendered LaTeX placeholders (' $\$$ ') that obscure the meaning. Additionally, the annotation on the red line contradicts the visual data, claiming a value increase that does not occur.

Figure 8

The figure displays a line chart of rollout generation time but is critically flawed by a non-existent caption that fails to describe the data, axes, or legend entries.

Figure 9

The figure contradicts its own caption by showing the 'EAGLE3 ours' line performing better (higher block efficiency) than the quantized baselines, whereas the text claims learned drafters remain below them. Additionally, the legend is overly dense and uses similar colors that reduce readability.

Figure 10

The figure displays a line plot of block efficiency but is rendered useless by a non-descriptive caption that fails to identify the data series or experimental conditions.

Figure 11

The figure clearly displays the performance comparison of rollout generation time across different methods, but the caption is merely a model name and lacks a descriptive summary of the data shown.

Figure 12

The figure clearly displays reward curves for two methods, but the caption is non-descriptive and the legend contains an undefined acronym ('AR') that obscures the baseline definition.

Action items.

- **[writing]** Figure 1: The caption is missing; the provided text ‘(no caption)’ does not describe the plot’s content, experimental setup, or the specific model/dataset used.
- **[writing]** Figure 1: The legend at the top is cut off, displaying only the line styles and percentages (50%, 90%, 99%) without the corresponding variable name or unit.
- **[writing]** Figure 3: The caption contains a placeholder ‘\$\$’ instead of the actual variable name (likely ‘gamma’ or ‘block size’), making the text incomplete.
- **[writing]** Figure 3: The y-axis label ‘Block Efficiency ()’ uses the symbol ‘ ’, but the caption refers to ‘block efficiency \$\$’, creating a notation mismatch.
- **[writing]** Figure 4: The caption contains raw LaTeX syntax (‘\$\$’) instead of the actual variable name (e.g., ‘ ’ or ‘gamma’), making the metric undefined in the text.
- **[writing]** Figure 4: The y-axis label ‘Block Efficiency (r)’ uses the symbol ‘r’, which contradicts the caption’s use of ‘\$\$’ and is not defined in the caption or legend.
- **[writing]** Figure 5: The figure lacks a formal caption; the text ‘Main figure-1-ver2’ appears to be a filename artifact rather than a descriptive caption.
- **[writing]** Figure 5: The legend at the top is not enclosed in a box or clearly separated from the diagram content, which may cause confusion regarding its scope.
- **[science]** Figure 6: The caption states the blue line marks $\text{Speedup}_{\{SD\}} = 1 + \epsilon$, but the legend explicitly defines it as $\text{Speedup} = 1 + \epsilon$ ($\epsilon = 0.05$). The caption fails to define the value of ϵ , making the threshold ambiguous without reading the legend.
- **[writing]** Figure 6: The y-axis label ‘Batch’ is ambiguous; it likely refers to ‘Batch Size’ given the range (0-60), but the unit is not specified.
- **[writing]** Figure 7: The caption contains unrendered LaTeX placeholders (‘\$\$’) instead of the variable name (likely ‘gamma’ or ‘draft length’), making the text unreadable.
- **[science]** Figure 7: The red line is annotated with ‘ 5.1 -> 5.6 (below thres)’, but the line is flat at $y=5$, contradicting the claim that the value increases to 5.6.
- **[writing]** Figure 7: The y-axis label ‘Draft Length ’ uses the Greek letter gamma, but the caption uses ‘\$\$’, creating a disconnect between the visual and the description.
- **[fatal]** Figure 8: The caption is non-existent (‘Qwen3-8B [filename]’), failing to describe the plot’s content, axes, or experimental conditions.
- **[science]** Figure 8: The legend lists ‘No-SD’ and ‘EAGLE3’ variants, but the caption does not define these terms or explain the experimental setup.
- **[writing]** Figure 8: The Y-axis label ‘Rollout Generation Time (s)’ is ambiguous; it is unclear if this represents total time, per-step time, or time per token.
- **[science]** Figure 9: The legend lists ‘Qwen2.5-7B (EAGLE3 ours)’ (solid blue) and ‘Qwen2.5-7B (EAGLE FastRL)’ (solid light blue), but the caption claims ‘evaluated learned auxiliary

drafters remain largely below target-induced quantized drafters.’ The solid blue line (EAGLE3 ours) is clearly ABOVE the dashed quantized lines for most of the plot, directly contradicting the caption’s claim.

- **[writing]** Figure 9: The legend is cluttered and difficult to read due to the high number of entries and similar line colors (e.g., multiple shades of blue and orange), making it hard to distinguish between specific model configurations like ‘Qwen2.5-7B (EAGLE3 ours)’ and ‘Qwen2.5-7B (EAGLE FastRL)’.
- **[fatal]** Figure 10: The caption is non-descriptive (‘Qwen2.5-7B’) and fails to explain the plot’s content, axes, or what the orange line represents, making the figure unintelligible.
- **[science]** Figure 10: The y-axis label ‘Block Efficiency (r)’ uses an undefined symbol ‘r’ without specifying units or the baseline for comparison.
- **[writing]** Figure 11: The caption ‘Qwen2.5-7B’ is insufficient; it fails to describe the plot’s content (Rollout Gen. Time vs. Training Step) or the specific methods being compared.
- **[writing]** Figure 11: The legend is placed outside the plot area at the top, which is non-standard and risks being cut off or misaligned in different renderings.
- **[writing]** Figure 12: The caption ‘Qwen2.5-7B’ is insufficient; it fails to describe the plot’s content (Reward vs. Training Step) or the specific methods being compared (‘veRL (AR)’ vs. ‘EfficientRollout’).
- **[science]** Figure 12: The legend uses the label ‘veRL (AR)’ without defining what ‘AR’ stands for (e.g., Autoregressive), which is necessary for interpreting the baseline comparison.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on domain-specific shorthand and undefined acronyms that create a barrier for readers outside the immediate sub-field of LLM inference optimization.

First, the core metric “block efficiency” (denoted as $\mathcal{B}$) is introduced in Section 3.1 without a plain-English definition. While the formula is provided, the text does not explicitly state that this represents the average number of accepted tokens per draft block, a concept crucial for understanding the speedup claims. Similarly, “regime-aware” (Section 3.2) is used as a key descriptor without initially defining the “regimes” (compute-bound vs. memory-bound) it toggles between, forcing the reader to infer the meaning from the Roofline model context.

Second, the paper assumes familiarity with specific quantization and hardware terminology. Acronyms such as “W4” (4-bit weights), “RTN” (Round-to-Nearest), and “AWQ” (Activation-aware Weight Quantization) appear in Section 3.1 and the Appendix without definition. While

standard in hardware-aware ML papers, these should be spelled out at first use (e.g., “4-bit weight quantization (W4) using Round-to-Nearest (RTN)”) to ensure accessibility. The term “TP=1” (Tensor Parallelism) in the Appendix is also used without expansion.

Finally, while “Speculative Decoding (SD)” is defined, the acronym “AR” (Autoregressive) is used repeatedly (e.g., “AR rollout baseline”) without an explicit definition. Given that the paper targets a broader audience interested in RL efficiency, defining these foundational terms would significantly improve readability without sacrificing technical precision.

Action items.

- **[writing]** The manuscript relies heavily on domain-specific shorthand and undefined acronyms that create a barrier for readers outside the immediate sub-field of LLM inference optimization. First, the core metric “block efficiency” (denoted as $\text{\mal}$) is introduced in Section 3.1 without a plain-English definition. While the formula is provided, the text does not explicitly state that this represents the average number of accepted tokens per draft block, a concept crucial for understanding the speedup claim.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a logically coherent framework for system-aware speculative decoding in RL rollouts, with a clear chain of reasoning from the identified bottlenecks (long-tail decoding, memory-bound regimes) to the proposed solutions (quantized self-drafter, regime-aware toggling, adaptive gamma). The premises regarding the inefficiency of static drafters in evolving RL policies and the shift from compute-bound to memory-bound regimes are well-supported by the cited literature and the provided empirical data (e.g., Fig 3, Table 1).

However, there are minor logical inconsistencies in the presentation of quantitative claims that require clarification:

1. **Latency Attribution (Sec 4.1):** The text states that “Dense projection time $\text{\Tdense}$ accounts for around 90% of total latency.” The supporting data in the Appendix (Table `decode_breakdown_qwen_llama`) shows FFN + QKVO contributions ranging from ~86% to ~89% depending on the model and batch size. While close, the jump to “90%” without specifying the exact regime or rounding convention slightly weakens the precision of the premise for the quantization strategy. A more precise statement or a specific reference to the worst-case regime would strengthen the logical link.
2. **Undefined Parameter (Sec 4.2):** The SD toggle policy is defined by the inequality $\frac{\text{\mal}, \widehat{\text{\TargetTime}}}{\text{\DraftLength} \widehat{\text{\DraftTime}} + \widehat{\text{\VerifyTime}}} \geq 1 + \text{\ToggleMargin}$. The variable $\text{\ToggleMargin}$ is introduced here but never assigned a numerical value in the main text or the algorithm description.

While the Appendix mentions $\epsilon=0.05$ for validation, the main logical flow of the algorithm relies on a parameter that is effectively hidden. For the logic to be fully self-contained and reproducible, the value of this margin should be explicitly stated in the method section.

3. **Comparative Claim Ambiguity (Sec 5.3):** The text claims the adaptive policy “achieves a 19.6% reduction in rollout-generation latency, compared to 13.5% and 11.8% for fixed $\gamma=5$ and $\gamma=11$.” The 19.6% figure in Table 1 is the reduction relative to the *No-SD* baseline. The 13.5% and 11.8% figures are also reductions relative to *No-SD*. The phrasing “compared to” implies a relative comparison between the adaptive method and the fixed methods (i.e., adaptive is X% better than fixed), but the numbers presented are all absolute improvements over the same baseline. The logic holds that adaptive is better, but the sentence structure conflates the absolute reduction with the comparative advantage, which could mislead a reader about the magnitude of the improvement *over* the fixed baselines.

These issues are primarily presentational and do not invalidate the core scientific logic, but they should be corrected to ensure the conclusions follow precisely from the stated premises and data.

Action items.

- **[writing]** Latency Attribution (Sec 4.1): The text states that “Dense projection time T_{dense} accounts for around 90% of total latency.” The supporting data in the Appendix (Table decode_breakdown_qwen_llama) shows FFN + QKVO contributions ranging from ~86% to ~89% depending on the model and batch size. While close, the jump to “90%” without specifying the exact regime or rounding convention slightly weakens the precision of the premise for the quantization strategy. A more precise statement or a specifi
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Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several claims that extend beyond the provided evidence, particularly regarding the preservation of model quality and the universality of the proposed solution’s advantages.

First, the abstract and introduction repeatedly claim that EfficientRollout preserves “final model quality” while reducing latency. However, the experimental section (Sec 5) and appendices (Figs 10-12) only present training curves (reward and validation accuracy) over a limited window of 100 steps. There is no data showing the final converged performance of the models trained with EfficientRollout versus the autoregressive baseline. Without reporting the final scores or conducting statistical significance tests on the converged models, the claim of “preserving final model quality” is an overreach. The observed similarity in early training dynamics does not guarantee identical final outcomes, especially given the stochastic nature of RL.

Second, the paper argues in Section 5.3 and Appendix D that learned auxiliary drafters are inherently difficult to obtain for RL workloads due to distribution mismatch. While the data supports that *publicly available* drafters (trained on ShareGPT) perform poorly, the conclusion that *any* learned drafter would fail is too strong. The authors do not attempt to train a custom drafter on the specific RL rollout distribution of their target models. A fair comparison would involve training a small auxiliary model on the rollout data generated by the target model itself. By only comparing against off-the-shelf drafters, the paper overstates the difficulty of the learned approach and overclaims the necessity of their self-speculative method.

Finally, the characterization of the method as “warm-up free” in Table 1 is misleading. While it avoids the multi-epoch pretraining required by learned drafters, it incurs a non-trivial “Prep.” time (1.3s to 2.6s per step in Table 2) for per-step quantization and requires a one-time calibration of the rooﬂine model parameters (Appendix A). This setup cost contradicts the implication of zero overhead or immediate applicability without any initialization phase. The text should be revised to accurately reflect these initialization costs rather than using the absolute term “warm-up free.”

Action items.

- **[science]** The claim that the method preserves ‘final model quality’ (Abstract, Intro) is overreaching. The paper only shows reward/accuracy curves over 100 steps (Figs 4, 10-12) without reporting final converged performance or statistical significance tests against the baseline.
- **[science]** The assertion that learned auxiliary drafters are ‘difficult to obtain’ due to distribution mismatch (Sec 5.3, App D) is partially unsupported. The paper compares against specific public drafters (ShareGPT-based) but does not rule out the possibility of training a custom drafter on the specific RL rollout distribution, which would be a fairer baseline.
- **[writing]** The claim of ‘warm-up free’ operation (Table 1) is exaggerated. The method requires a per-step quantization overhead (1.3s-2.6s prep time in Table 2) and a calibration phase for the rooﬂine model (App A). This is not truly ‘warm-up free’ in the same sense as history-based methods that require no setup.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript presents a system-aware speculative decoding framework for RL rollouts. From a safety and ethics perspective, the primary concerns relate to data provenance and the potential for accelerated generation of harmful content.

Data Privacy and Consent: The experimental setup (Section 5.1, Appendix B) relies on external datasets, specifically ShareGPT and SimpleRL. While these are public, the manuscript does not explicitly state whether the authors verified that the data was collected with user consent suitable for secondary research or if it complies with relevant privacy regulations (e.g., GDPR, CCPA). Given the increasing scrutiny on training data sources, a statement confirming that the datasets were used in accordance with their licenses and that no personally identifiable information (PII) was retained or exposed is necessary. Please add a “Data Ethics” or “Compliance” subsection in the Experimental Setup or Appendix.

Dual-Use and Harmful Content Generation: The proposed method significantly accelerates the generation of reasoning traces (rollouts) by up to ~20% (Table 1). While the paper demonstrates that model quality (reward/accuracy) is preserved, it does not address whether the acceleration could inadvertently increase the throughput of harmful, biased, or unsafe reasoning patterns during the training phase. RL training often involves exploring the action space; faster rollouts mean more exploration steps per unit time. The authors should briefly discuss whether their safety filters or reward models are robust enough to handle this increased throughput or if the acceleration introduces new risks in the deployment of the final model. A short paragraph in the Discussion (Section 6) addressing these potential downstream safety implications is recommended.

Conflicts of Interest: The Acknowledgments section mentions support from the “Advanced GPU Utilization Support Program” and the “FuriosaAI team.” While this is disclosed, the paper should explicitly confirm if any authors have financial ties to FuriosaAI or if the hardware used was provided under conditions that might bias the results. A standard conflict of interest statement is advisable.

No fatal ethical violations were found, but the lack of explicit data compliance statements and safety impact analysis prevents an immediate acceptance.

Action items.

- **[writing]** The paper uses datasets like ShareGPT and SimpleRL without explicit confirmation of user consent or IRB approval for secondary research use. Add a statement in Section 5.1 (Setup) or Appendix B confirming compliance with data privacy regulations and the terms of service for all external datasets.
- **[writing]** The method involves generating long-form reasoning traces (rollouts) for RL training. While the paper claims quality preservation, it lacks a specific discussion on

potential safety risks (e.g., generating harmful reasoning paths) during the accelerated rollout phase. Include a brief risk assessment or mitigation strategy in Section 6 (Discussion).

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a system-aware speculative decoding framework for RL rollouts, but the scientific evidence supporting the central claims requires strengthening in three key areas: statistical rigor, hyperparameter sensitivity, and baseline control.

First, the claim that EfficientRollout “preserves final model quality” (Section 5.2) is supported primarily by visual overlays of training reward curves (Figure 3c, Appendix Figure 4). While the curves appear similar, the paper reports only 100 training steps. Without statistical significance testing (e.g., t-tests or confidence intervals) on the final validation accuracy or reward scores, it is impossible to rule out that the observed differences are due to random variance rather than true equivalence. The authors should report the mean and standard deviation of the final metrics across multiple seeds (if available) or at least provide statistical tests comparing the final performance of EfficientRollout against the No-SD baseline.

Second, the adaptive draft-length policy (Algorithm 1) relies on specific hyperparameters: a patience window $P=2$ and acceptance thresholds $\alpha_{\text{up}}=0.94$, $\alpha_{\text{down}}=0.85$. The results indicate that for Llama3.1-8B, the policy never triggered an increase in γ , remaining static at 5. The paper does not provide a sensitivity analysis to demonstrate that these specific thresholds are robust. If the thresholds were slightly different, would the policy have adapted? A robustness check varying these hyperparameters (e.g., ± 0.05) or a justification for their selection based on the observed entropy dynamics would strengthen the evidence that the adaptive mechanism is effective and not just a tuned artifact for Qwen models.

Third, the comparison against “Learned auxiliary” baselines (EAGLE3) reveals a large disparity in block efficiency (τ) between models (e.g., $\tau \approx 1.2$ for Llama vs. $\tau \approx 3.6$ for Qwen). The authors attribute this to “training-distribution mismatch” (Appendix Section 4.5). However, the evidence for this is qualitative. The paper does not quantify the statistical significance of the performance gap between the self-drafter and the learned baselines, nor does it fully control for the pretraining data of the external drafters. For instance, the Llama baseline uses a RedHatAI drafter, while Qwen uses a ShareGPT-pretrained one. The difference in pretraining data quality or domain alignment could be a confounding variable. The authors should either retrain the learned baselines on the exact same RL rollout data to isolate the “online adaptation” factor or provide a more rigorous statistical analysis of the distribution mismatch hypothesis.

Finally, the sample size for the “Regime-aware SD toggle” validation (Appendix Section 2) relies on a single A100 GPU with TP=1. While the roofline model is theoretically sound, the

empirical validation of the toggle boundary (Figure 5) should ideally be repeated across different hardware configurations or batch sizes to ensure the “conservative” policy generalizes beyond the specific calibration setup.

Action items.

- **[science]** The claim of ‘preserved final model quality’ relies on visual inspection of reward curves (Fig. 3c, App. Fig. 4) without statistical validation. Add p-values or confidence intervals for the final reward/accuracy differences between EfficientRollout and the No-SD baseline to rule out random variance, especially given the small number of training steps (100) reported.
- **[science]** The adaptive draft-length policy (Alg. 1) uses a ‘patience window P’ and thresholds (alpha_up/down) but lacks a sensitivity analysis. The results show Llama3.1-8B never adjusted gamma. Provide a robustness check showing how performance varies if these hyperparameters are perturbed, or justify why the specific values chosen are optimal across different model scales.
- **[science]** The comparison against ‘Learned auxiliary’ baselines (EAGLE3) shows significant variance in block efficiency across models. The paper attributes this to distribution mismatch but does not quantify the statistical significance of this difference or control for the specific pretraining data size/quality of the external drafters used. Clarify if the baselines were retrained on the exact same RL data or if the comparison is confounded by pretraining data differences.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The manuscript presents a system-aware speculative decoding framework with compelling latency reductions. However, the statistical rigor of the experimental evaluation requires strengthening to support the claims of robustness and reproducibility.

1. Lack of Variance and Significance Testing In Table 1 (Section 5.1), the authors report single-point latency reductions (e.g., -19.6% for Qwen2.5-7B rollout generation) without any measure of variance (standard deviation, standard error) or statistical significance testing. Given that RL training dynamics can be stochastic, a single run or a single aggregated average is insufficient to claim a definitive speedup. The authors should report results averaged over multiple independent training seeds (e.g., $N \geq 3$) with error bars or confidence intervals. Without this, it is impossible to determine if the observed improvements are statistically distinguishable from noise or baseline fluctuations.

2. Correlation Analysis Details Section 4.3 states a strong negative correlation ($r = -0.99$) between target-policy entropy and quantized-drafter acceptance. While the magnitude suggests

a strong relationship, the statistical context is missing. The text does not specify the sample size (\$N\$) used to calculate this correlation coefficient (e.g., number of rollout steps or tokens). A correlation of -0.99 with a small \$N\$ is less reliable than with a large \$N\$. Additionally, a hypothesis test (e.g., t-test for correlation) should be provided to confirm the significance of this relationship, rather than just reporting the coefficient.

3. Aggregation Methodology Appendix E001 mentions that timing metrics are “averages over the subsequent 100 steps.” It is unclear if these 100 steps are contiguous or sampled, and whether the “100 steps” represents the entire evaluation window or a subset. Furthermore, the method of aggregation (mean vs. median) is not explicitly defined for the final reported numbers in Table 1. In systems experiments where outliers (e.g., GC pauses, network jitter) can skew means, reporting the median with interquartile ranges is often more appropriate. The authors should clarify the aggregation strategy and provide the distribution of the underlying data points.

4. Baseline Consistency The comparison against baselines (e.g., `\alwaysd{}`) shows mixed results (e.g., `\alwaysd{}` is slower on Qwen2.5-14B). While the paper attributes this to regime dynamics, the statistical stability of these negative results is not quantified. If the variance in the baseline is high, the claim that `\methodtitle{}` is “consistently” better needs stronger statistical backing.

To address these concerns, the authors should re-run experiments with multiple seeds, report mean $\pm$ standard deviation (or 95% confidence intervals) for all latency metrics, and provide the sample size and significance test for the correlation analysis.

Action items.

- **[science]** Report statistical significance (e.g., p-values, confidence intervals, or standard deviations) for the latency reductions in Table 1. The current presentation of single-point averages (e.g., -19.6% for Qwen2.5-7B) lacks evidence of reproducibility across training runs or seeds.
- **[science]** Clarify the statistical basis for the correlation claim ($r=-0.99$) in Section 4.3. Specify the sample size (number of steps or tokens) used to compute this Pearson correlation and whether it was tested for significance.
- **[science]** Define the aggregation method for the ‘100 steps’ mentioned in Appendix E001. Explicitly state if these are consecutive steps, if variance was calculated, and whether the reported metrics are means with standard errors or medians with interquartile ranges.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a technically dense contribution, but the writing quality is currently

compromised by an over-reliance on undefined LaTeX macros and inconsistent notation, which significantly hinders readability.

The most critical issue is the extensive use of custom macros such as `\methodtitle`, `\bottleneck`, `\speeduprollout`, `\speedupete`, `\nosd`, `\history`, `\learned`, and `\alwaysd`. In the provided source, these commands are not expanded, rendering sentences like “We present `\textbf{\methodtitle{}}...`” and “compared to `\nosd{}` (AR)” opaque to a reader without the specific preamble. While common in draft stages, a final manuscript must either define these macros clearly in the preamble or, preferably, replace them with their full text or standard formatting to ensure the prose stands on its own. For instance, “EfficientRollout” should be written out, and “No-SD” should be used instead of `\nosd`.

Furthermore, the mathematical notation in the Preliminary and Method sections lacks consistency. Symbols like `\mal` (block efficiency) and `\TargetTimebs` are introduced abruptly. While the context suggests their meaning, a formal paper should define these variables explicitly upon first use or provide a nomenclature list. The transition between text and equations is sometimes abrupt, as seen in Section 3.2 where the toggle policy equation is presented with minimal textual scaffolding explaining the specific terms within the inequality.

Finally, there is minor inconsistency in citation formatting, with a mix of `\cite`, `\citep`, and `\citet` used throughout the text. Standardizing these to a single style (likely `\citep` for parenthetical citations) would improve the visual flow. Addressing these macro and notation issues is essential to make the paper accessible to a broader audience beyond those familiar with the authors’ specific LaTeX template.

Action items.

- **[writing]** Replace all custom LaTeX macros (e.g., `\methodtitle`, `\bottleneck`, `\speeduprollout`) with their expanded text equivalents or standard formatting commands. The current manuscript relies on undefined macros that obscure the prose and prevent immediate readability without the preamble.
- **[writing]** Clarify the variable definitions in Section 3.1 and 3.2. The text introduces symbols like `\mal`, `\TargetTimebs`, and `\SDBlockTime` without a consistent nomenclature table or immediate definition, forcing the reader to guess their meaning from context.
- **[writing]** Standardize the citation style. The manuscript inconsistently uses `\cite`, `\citep`, and `\citet`. Ensure a uniform style (e.g., all parenthetical) is applied throughout the text to maintain professional flow.